

Testing the Fluorescence Advantage: Solar-Induced Fluorescence as an Early Indicator of Agricultural Drought Stress in Marathwada, Maharashtra

Abstract

Drought detection frameworks used for official drought declaration and agricultural insurance assessment in India rely substantially on rainfall-deficit records and the Normalized Difference Vegetation Index (NDVI), both of which respond to crop stress only after visible physiological degradation has already occurred. This study examines whether Solar-Induced Fluorescence (SIF), a satellite-derived proxy for photosynthetic activity, registers vegetation stress measurably earlier than NDVI during documented drought conditions. Using GOSIF v2 fluorescence data and cloud-screened MODIS NDVI over the Marathwada region of Maharashtra across two drought years (2015, 2018) and one normal monsoon year (2020), a quantitative lag was calculated between the two indices' post-peak seasonal decline. SIF's decline preceded NDVI's decline consistently across all three years studied — a result independently corroborated by a second, methodologically distinct cross-correlation lag estimate — but the hypothesis that drought conditions amplify this lag was not supported: mean lag was smaller in the two drought years (15.6 days) than in the normal year (28.7 days), and the two drought years differed substantially from each other (24.3 versus 6.9 days). The precise inter-annual ranking of lag magnitude is itself method-sensitive, and is reported with that caveat. Independent validation using CHIRPS precipitation data confirmed the drought/normal classification (rainfall departures of -21.5% , -18.3% , and $+29.1\%$ from a twenty-year climatological normal, respectively) and revealed that the 2018 drought was spatially concentrated in the western and southern districts, while eastern districts recorded near-normal or surplus rainfall. District-level spatial analysis confirms this pattern statistically: mean SIF and rainfall anomaly are strongly and significantly correlated across districts and years (Pearson $r = 0.837$, $p < 0.001$; Spearman $\rho = 0.857$, $p < 0.001$), providing a physically coherent and quantitatively supported cross-validation between the two independently derived satellite datasets. Three supplementary analyses were added following external review of this study. A case-resampling bootstrap (2,000 replicates per year) confirms that the *direction* of the SIF-leads-NDVI finding is robust (SIF's estimated lag was non-negative in 100% of replicates for every year) while showing that the *exact magnitude* carries wide uncertainty at this sample size, with every pairwise between-year comparison statistically indistinguishable. Moran's I confirms, quantitatively, that the district-level correlation above is drawn from spatially autocorrelated rather than independent observations (all three years significant at $p < 0.05$ for both variables), consistent with this paper's existing caveat about effective sample size. A supplementary comparison against the one well-documented official government drought-declaration date available (Maharashtra, 31 October 2018) found both SIF and NDVI crossed their decline thresholds roughly seven weeks earlier, with SIF's specific edge over NDVI (3–4 days) small relative to that larger gap.

Keywords: solar-induced fluorescence, remote sensing, drought monitoring, NDVI, agricultural stress, Marathwada

1. Introduction

1.1 Motivation

The Normalized Difference Vegetation Index is a reflectance-based measure: it quantifies how much visible and near-infrared light a plant canopy reflects, and this reflectance changes only once a plant's internal structure has already begun to visibly degrade — thinning leaves, reduced chlorophyll content, canopy stress. Solar-Induced Fluorescence is grounded in a more direct physical process. When a leaf absorbs sunlight for photosynthesis, a small fraction of the absorbed energy is not converted into chemical energy but is instead re-emitted as fluorescence, governed by the quantum yield of photosynthesis. Under physiological stress, photosynthetic efficiency declines before outward greenness does, meaning SIF is expected to register stress earlier than NDVI. Where NDVI reflects a plant's outward appearance, SIF reflects what is occurring physiologically within it.

India's drought declaration process, and the crop-insurance payout mechanism under the Pradhan Mantri Fasal Bima Yojana (PMFBY) that depends on it, relies substantially on rainfall-deficit and NDVI-based assessment. Both indicators are known to register drought only after visible crop stress has already set in, and this delay translates directly into delayed insurance payouts at precisely the point when affected farmers require them most urgently. This study treats the existence and size of a SIF–NDVI lag as an empirical, testable question, with the aim of establishing whether such a lag constitutes a physics-based case for an earlier-warning indicator than those presently used in policy.

1.2 Physical Basis of Solar-Induced Fluorescence

When a leaf absorbs photosynthetically active radiation, the absorbed energy is partitioned among three competing, mutually exclusive pathways at the chlorophyll level: photochemistry (Φ_P), non-photochemical quenching or heat dissipation (Φ_{NPQ}), and chlorophyll fluorescence (Φ_F). These three pathways account for the entirety of absorbed energy, described by the conservation relationship:

$$\Phi_P + \Phi_{NPQ} + \Phi_F = 1$$

Under well-watered, physiologically normal conditions, a large share of absorbed energy is directed toward photochemistry, with a comparatively small and relatively stable fraction re-emitted as fluorescence. Under drought or heat stress, the photochemical pathway is down-regulated as the photosynthetic apparatus protects itself from damage, while the non-photochemical quenching pathway is up-regulated to dissipate the resulting surplus energy as heat. This re-partitioning alters the fluorescence yield in a manner that is measurable before any outward change in canopy structure or greenness occurs, which is the physical basis for SIF's expected earlier-warning capacity relative to reflectance-based indices such as NDVI (Figure 1).

The magnitude of the fluorescence signal actually detected by a satellite sensor, denoted SIF, is the product of three quantities:

$$\text{SIF} = \text{APAR} \times \Phi_F \times f_{\text{esc}}$$

where APAR is the absorbed photosynthetically active radiation, Φ_F is the fluorescence quantum yield described above, and f_{esc} is the fraction of emitted fluorescence photons that escape the canopy without being reabsorbed. Because chlorophyll fluorescence is several orders of magnitude weaker than reflected sunlight at the same wavelengths, it cannot be measured through simple radiance detection. Satellite-based SIF retrievals, as performed by instruments such as GOME-2, OCO-2, and TROPOMI, instead exploit narrow, naturally dark absorption features in the solar spectrum — Fraunhofer lines — where the additional radiance contributed by fluorescence emission can be isolated from the reflected background. The GOSIF product used in this study is not itself a direct satellite retrieval, but a statistically modeled reconstruction combining

discrete OCO-2 SIF soundings with continuous MODIS reflectance data and meteorological reanalysis to produce spatially and temporally continuous global coverage.

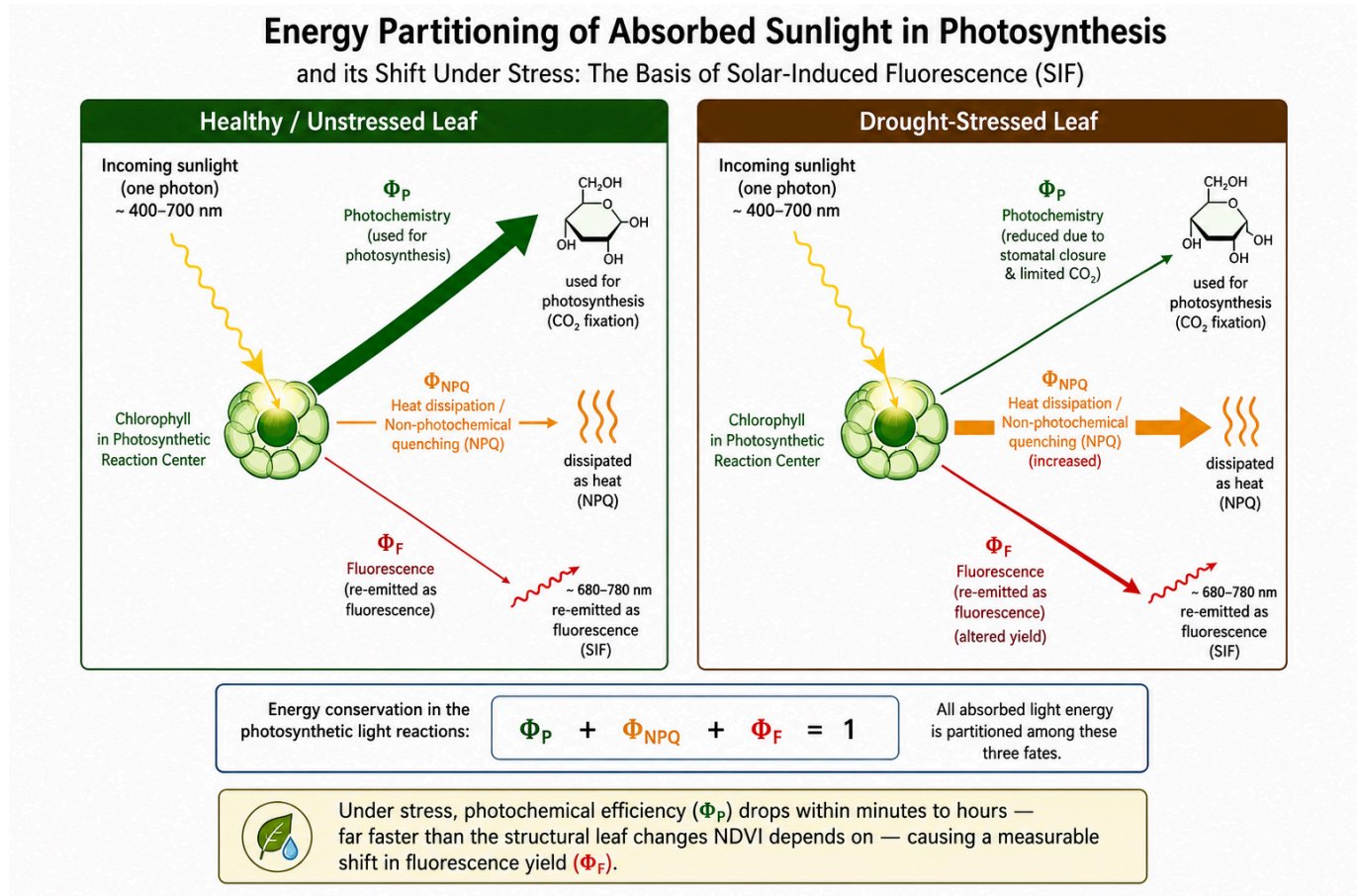


Figure 1. Schematic representation of leaf-level light energy partitioning among photochemistry, non-photochemical quenching (heat dissipation), and chlorophyll fluorescence, under well-watered versus drought-stressed conditions.

1.3 Physical Basis of NDVI

The Normalized Difference Vegetation Index is computed from surface reflectance in the red and near-infrared bands:

$$\text{NDVI} = (\text{NIR} - \text{RED}) / (\text{NIR} + \text{RED})$$

This formulation exploits a specific optical property of healthy plant tissue. In the red portion of the spectrum, chlorophyll pigments within the leaf's palisade mesophyll strongly absorb incoming light to drive photosynthesis, resulting in low red reflectance. In the near-infrared, by contrast, the internal leaf structure — specifically the irregular air-cell interfaces within the spongy mesophyll layer — causes strong multiple scattering due to the refractive-index mismatch between cell walls and intercellular air spaces, resulting in high near-infrared reflectance that is largely independent of chlorophyll content (Figure 2). A canopy with abundant, healthy, chlorophyll-rich foliage therefore produces a large contrast between low red and high near-infrared reflectance, yielding a high NDVI value; a canopy undergoing structural degradation — thinning leaves, reduced leaf area, senescence — reduces this contrast and lowers NDVI.

Because this signal depends on the plant's outward structural and pigment state, NDVI necessarily lags behind the internal physiological changes that precede visible degradation. Cloud and cloud-shadow contamination, arising from Mie scattering, further complicates the NDVI signal in the optical domain, an issue explicitly

addressed in this study via MOD13Q1's [SummaryQA](#) per-pixel quality flag prior to any analysis (Section 3.1). This structural, appearance-based basis for NDVI stands in direct physical contrast to the internal-process basis of SIF described in Section 1.2, and this contrast is the central physical premise investigated throughout this study.

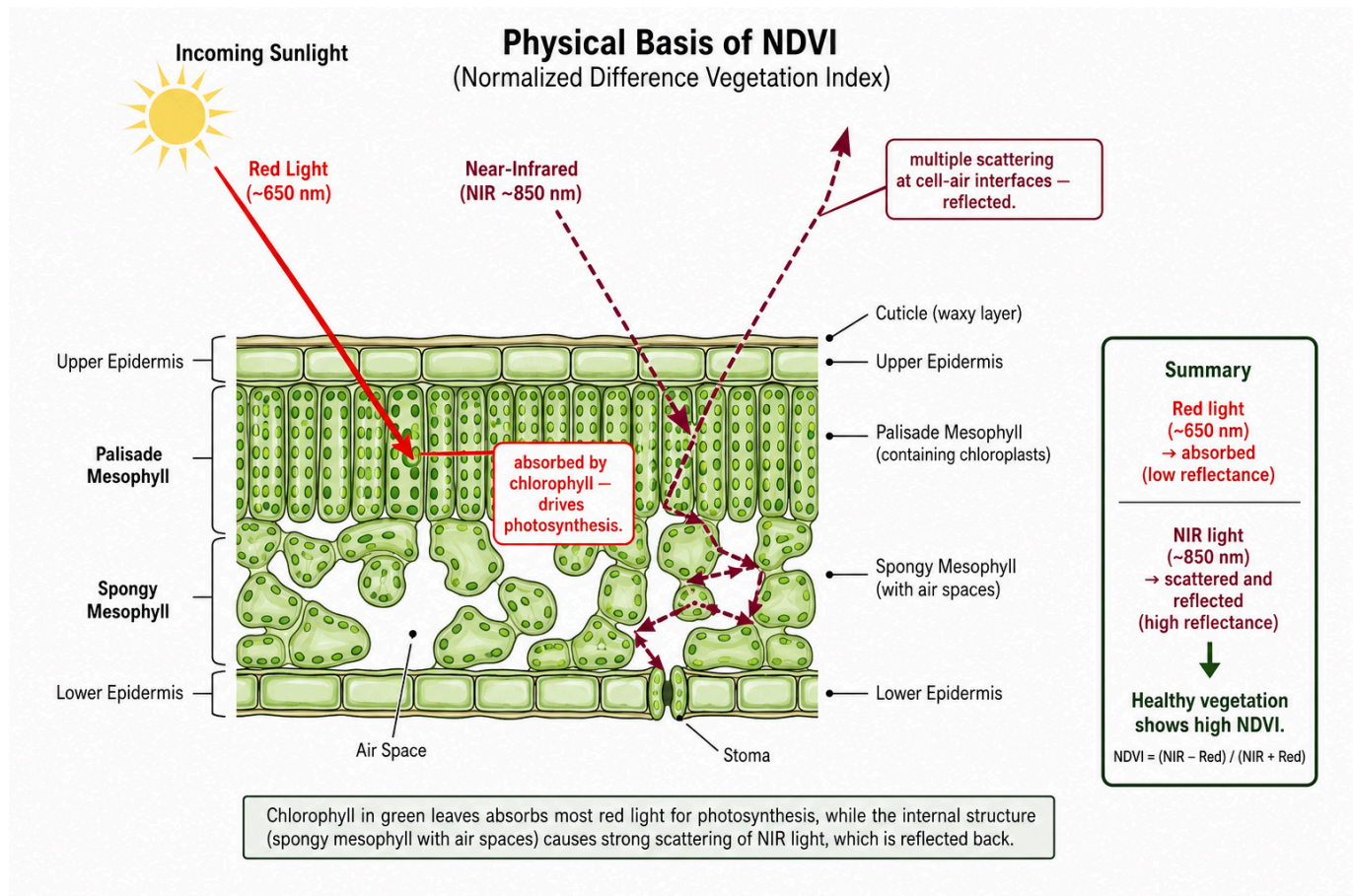


Figure 2. Leaf cross-section illustrating red-light absorption by chlorophyll in the palisade mesophyll versus near-infrared scattering at spongy mesophyll air-cell interfaces, the physical basis of the reflectance contrast measured by NDVI.

1.4 Research Questions

RQ1: Does SIF decline measurably earlier than NDVI during documented drought-onset periods in the study region?

RQ2: Is the SIF–NDVI lag, where present, consistent across the study region, or does it vary meaningfully by geography?

RQ3: How does the timing of SIF-based stress onset relate to independently measured rainfall deficit across drought and normal years?

1.5 Hypotheses

H1: SIF declines significantly before NDVI during a given drought episode.

H2: The magnitude of SIF–NDVI divergence is not spatially uniform across the study region.

H3: Drought severity, measured independently through rainfall deficit, amplifies the magnitude of the SIF–NDVI lag.

2. Study Area

The study region comprises the eight districts constituting Marathwada, Maharashtra — Chhatrapati Sambhajnagar (Aurangabad), Jalna, Parbhani, Hingoli, Nanded, Beed, Latur, and Dharashiv (Osmanabad) — selected for its recurring, well-documented history of agricultural drought.

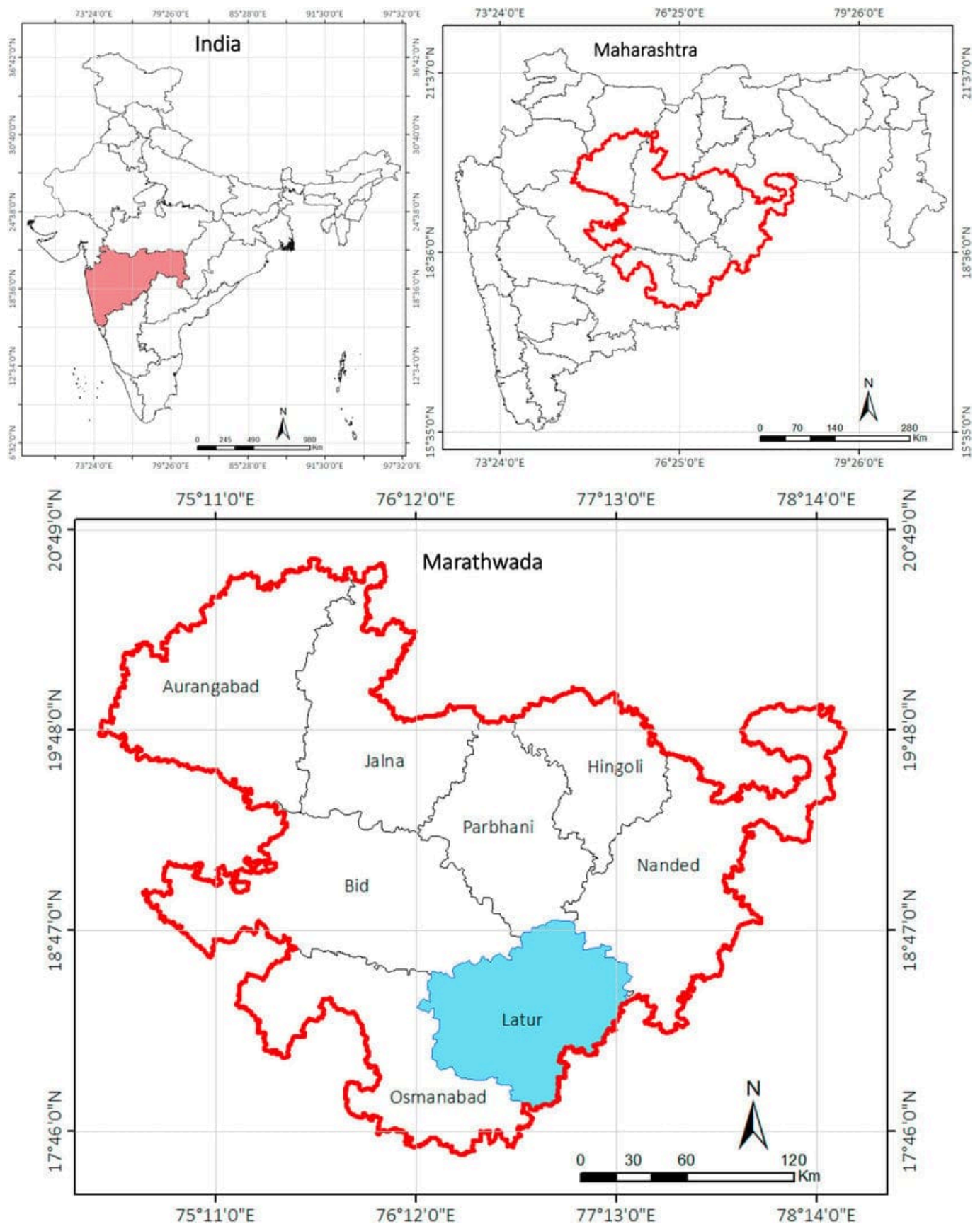


Figure 3. Location of the study area. (a) Maharashtra state within India. (b) Marathwada division within Maharashtra. (c) The eight constituent districts of Marathwada, with Latur district highlighted for reference.

3. Data and Methodology

3.1 Data Sources

Four datasets were used, all covering the June 1 – December 31 window for 2015, 2018, and 2020:

- **SIF:** GOSIF v2, a global 0.05°, 8-day solar-induced fluorescence product (Li & Xiao, 2019), distributed by the Global Ecology Group, University of New Hampshire.
- **NDVI:** MODIS MOD13Q1, a 16-day, 250 m Vegetation Index product with an internal Maximum-Value-Composite cloud-suppression algorithm, accessed via Google Earth Engine.
- **Land cover:** MODIS MCD12Q1 (IGBP classification), used to derive a cropland mask (classes 12, 14).
- **Precipitation:** CHIRPS daily rainfall (UCSB-CHG), accessed via Google Earth Engine, for the three study years and for a twenty-year (2001–2020) climatological baseline.

3.2 Boundary Definition and Verification

The study region boundary was defined using the FAO GAUL 2015 level-2 administrative dataset, filtered to the eight Marathwada districts and merged into a single precise polygon. This replaced an initial rectangular bounding-box definition, which was found on visual inspection to extend beyond Marathwada into neighboring Telangana districts and into Solapur and Yavatmal — areas outside the study region with distinct rainfall regimes. Both the SIF raster-clipping pipeline (using polygon-based masking) and the NDVI extraction pipeline (using the same polygon as the reducer geometry) were built against this corrected boundary.

Following this correction, the masking behavior was independently verified by overlaying the true boundary polygon directly on a sample clipped SIF raster, confirming that data was retained only within the actual district outlines and excluded elsewhere.

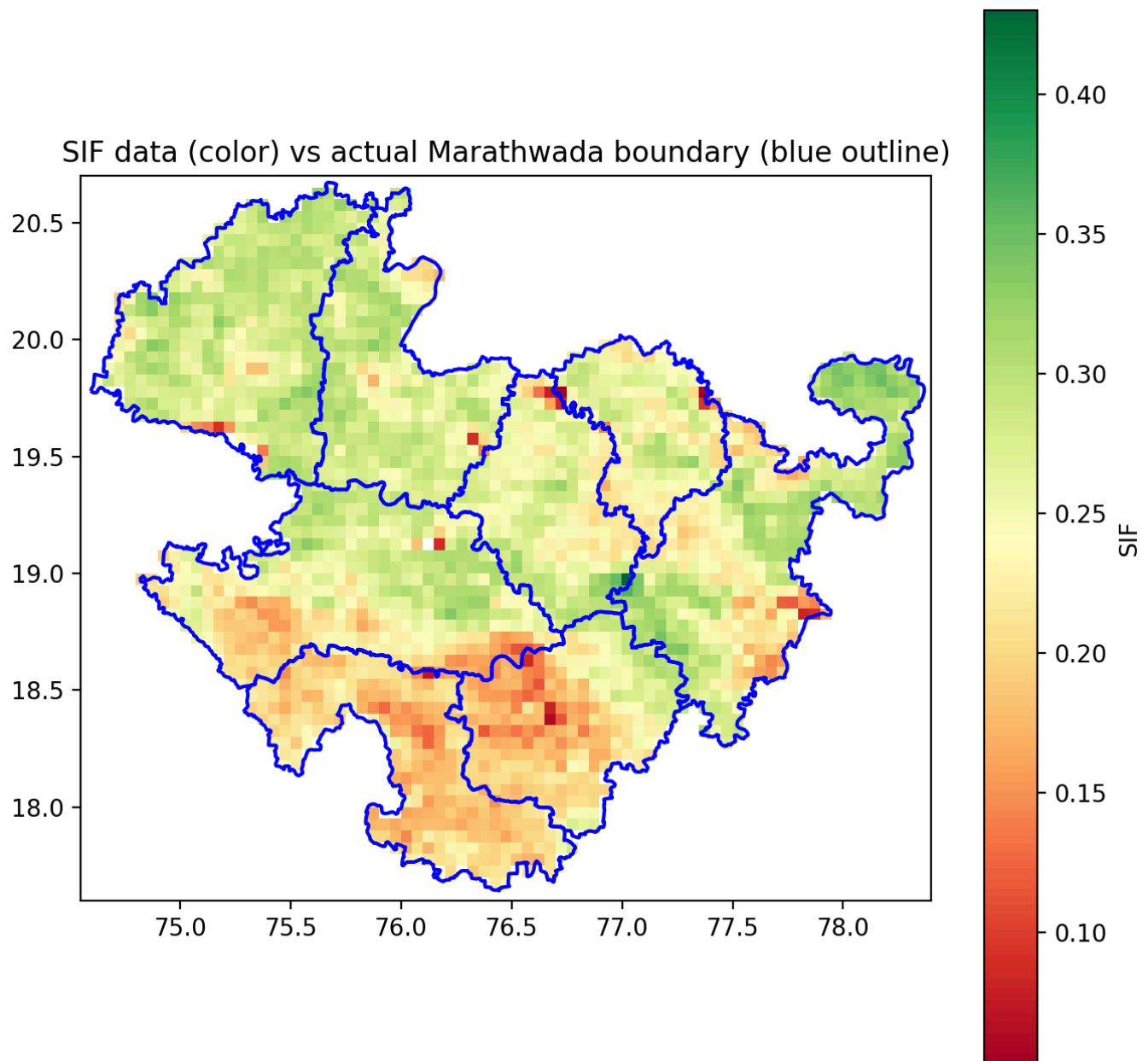


Figure 4. Verification of boundary masking: clipped SIF data (color) overlaid with the true Marathwada district boundary (blue outline), confirming that raster values are correctly excluded outside the study region.

3.3 Processing Pipeline

SIF GeoTIFFs were clipped to the study boundary, scaled by GOSIF's 0.0001 factor, and masked for fill-value codes (32766, water; 32767, non-vegetated/missing) prior to averaging. NDVI was extracted using MOD13Q1's *SummaryQA* band (retaining good- and marginal-quality pixels only) combined with the cropland mask. The two series, at different native temporal resolutions (8-day SIF, 16-day NDVI), were aligned using nearest-date matching within each year.

3.4 Lag Calculation

Each year's SIF and NDVI series were normalized (0–1) within-year, and the day-of-year at which each series crossed a set of decline thresholds (90%, 80%, 70%, 60%, 50% of seasonal peak) was estimated via linear interpolation between observed dates. Lag was defined as the difference between the NDVI and SIF crossing dates at each threshold.

3.5 Rainfall Validation

CHIRPS-derived seasonal rainfall totals were computed for the three study years and compared against a twenty-year (2001–2020) climatological mean and standard deviation for the same region and seasonal window, yielding a percentage anomaly and z-score for each study year. This was extended to the district level using the same regional climatological baseline as a shared reference across all eight districts.

3.6 Cross-Correlation Robustness Check

The threshold-crossing method (Section 3.4) estimates lag from when each series crosses a given percentage of its own seasonal peak — a measure most sensitive to the *onset* of decline, particularly at high thresholds. As an independent check on whether this specific choice of method drives the reported lag values, a second, methodologically distinct lag estimate was computed using time-lagged cross-correlation, which instead asks what single time-shift best aligns the two curves' overall shape across the full post-peak decline window. For each year, both series (normalized 0–1, restricted to the post-peak decline phase from SIF's own peak onward) were interpolated onto a common daily grid, de-measured, and cross-correlated at lags from 0 to $N/4$ days, where N is the length of the decline window in days — a standard bound that avoids the unreliable, boundary-pinned estimates that result from testing lags approaching the full series length. The lag maximizing the correlation between $SIF(t)$ and $NDVI(t + lag)$ was taken as the cross-correlation-based lag estimate for that year.

3.7 Uncertainty Quantification (Bootstrap Confidence Intervals)

The cross-correlation lag estimates in Section 3.6 are point estimates computed from a small number of discrete satellite observations per year — 17 post-peak, SIF/NDVI-matched 8-day observations in each year's decline window. To quantify how much these estimates could plausibly vary given the sparseness of the underlying observation dates, each year's lag estimate was bootstrapped using case resampling: the original 17 (day-of-year, SIF, NDVI) observations were resampled with replacement 2,000 times, each replicate was re-interpolated onto a daily grid and re-processed through the identical cross-correlation procedure described in Section 3.6, and the 2.5th and 97.5th percentiles of the resulting lag distribution were taken as a 95% confidence interval around the original point estimate. An earlier version of this analysis used a moving-block bootstrap applied directly to the interpolated daily series; that approach is documented, and abandoned, in Development_Log.md Entry 13, because block-shuffling a strongly non-stationary decline curve destroys the trend that produces the lag signal in the first place, collapsing every year's interval to a spurious single point at lag 0 rather than measuring genuine sampling uncertainty. The case-resampling approach used here instead resamples which underlying satellite overpass dates are represented in the fit — the more realistic source of uncertainty given an 8-day, cloud-gap-affected revisit cycle — while preserving the shape of the decline curve in every replicate.

3.8 Spatial Autocorrelation Diagnostic

The district-level Pearson/Spearman correlation reported in Section 4.5 already carries the caveat that its 24 district-year observations are not fully independent, since the eight districts are geographically adjacent and share regional weather systems. To move beyond a qualitative caveat, Moran's I — a standard measure of spatial autocorrelation — was computed for both mean SIF and rainfall anomaly, separately for each study year, using a Queen-contiguity spatial weights matrix built directly from the FAO GAUL district polygons used throughout this study (mean of 3.25 neighboring districts per district), with significance assessed via a 9,999-permutation test.

3.9 Comparison Against the Official Drought-Declaration Timeline (RQ3)

RQ1 (Section 1.4) asks whether SIF leads NDVI; Sections 4.1–4.2 and 4.6 answer this directly. RQ3, as originally posed in this project's planning documents, asks how satellite-detected stress onset relates to independently measured rainfall deficit — a version already answered by Sections 3.5 and 4.4–4.5. A related, more policy-relevant question — how satellite-detected stress onset compares to the timing of the *official* government drought declaration that governs PMFBY payout timing — was part of this project's early research-question list (Project_Journal.md) but was not addressed in earlier drafts of this paper, a gap identified during external review of this project. This section closes that gap for the one study year with a single, well-documented official declaration date: the Maharashtra government's Kharif drought declaration of 31 October 2018 (day-of-year 304), covering 151 talukas across 26 districts, including all eight Marathwada study districts (Zee News, 2018; Economic and Political Weekly, 2018). A comparably well-documented, single declaration date specific to 2015 could not be located during this review; that year's Marathwada drought is extensively documented in the context of the subsequent, better-known 2016 water-scarcity crisis rather than as a single dated 2015 Kharif declaration, so this comparison is reported for 2018 only rather than extrapolated to a year where the corresponding public record was not verifiable.

4. Results

4.1 Seasonal SIF–NDVI Trajectories

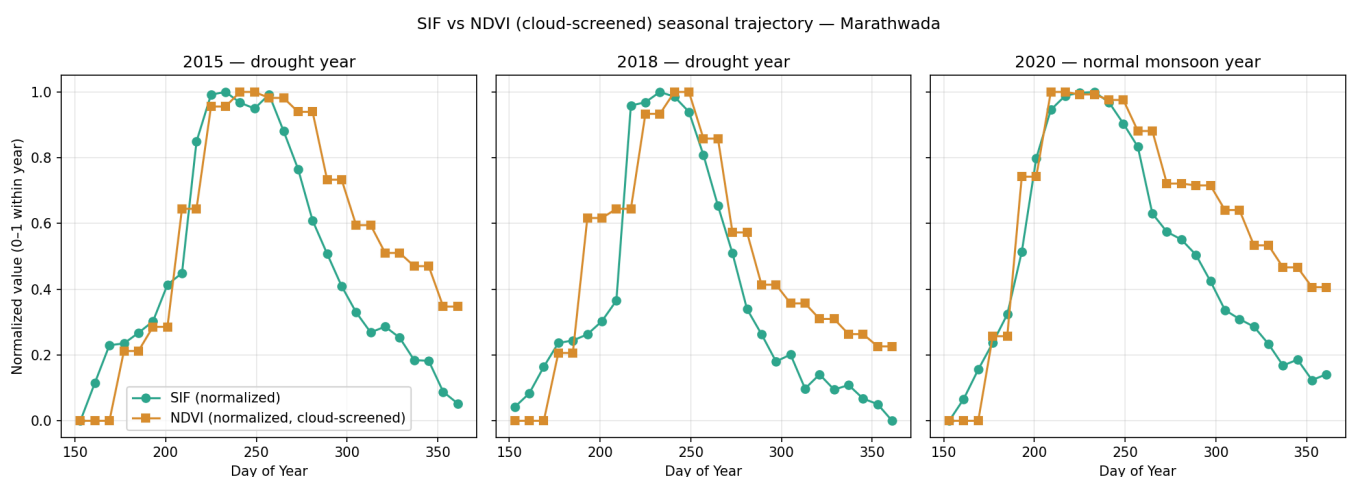


Figure 5. Normalized SIF and NDVI seasonal trajectories, Marathwada, 2015 (drought), 2018 (drought), and 2020 (normal monsoon). SIF's decline from seasonal peak precedes NDVI's decline in all three years.

Across all three years, SIF began its post-peak decline before NDVI, with NDVI remaining near its peak value for a longer period after SIF had already started dropping. This pattern was consistent regardless of drought classification.

4.2 Quantitative Lag Analysis

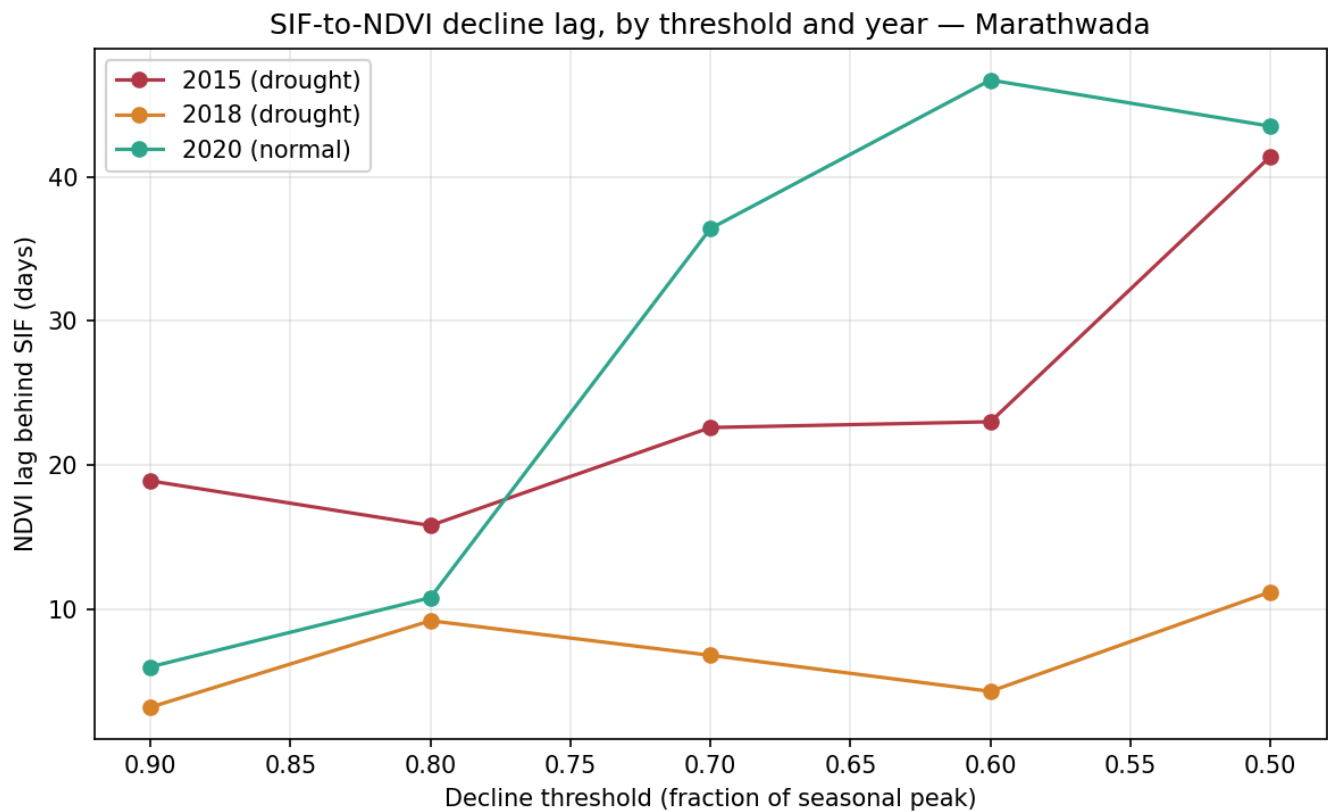


Figure 6. SIF-to-NDVI decline lag (days), by decline threshold and year.

Mean lag across all five resolved thresholds was 24.3 days (2015), 6.9 days (2018), and 28.7 days (2020). Grouped by drought classification, the two drought years averaged a smaller lag (15.6 days) than the normal year (28.7 days) — the opposite of the direction predicted by H3. The two drought years also differed substantially from one another (24.3 versus 6.9 days), a difference larger than either year's difference from the normal year.

4.3 Spatial Analysis — SIF

Solar-Induced Fluorescence — Marathwada (precise boundary) — DOY 273

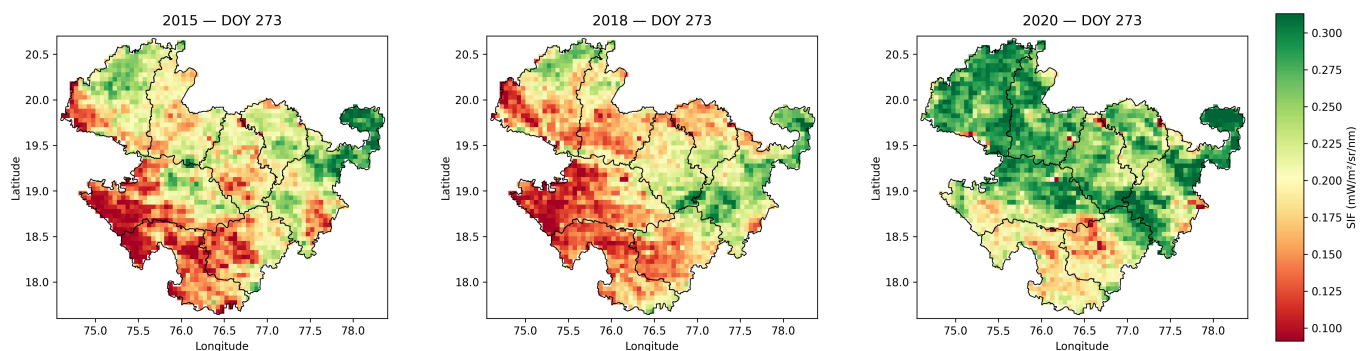
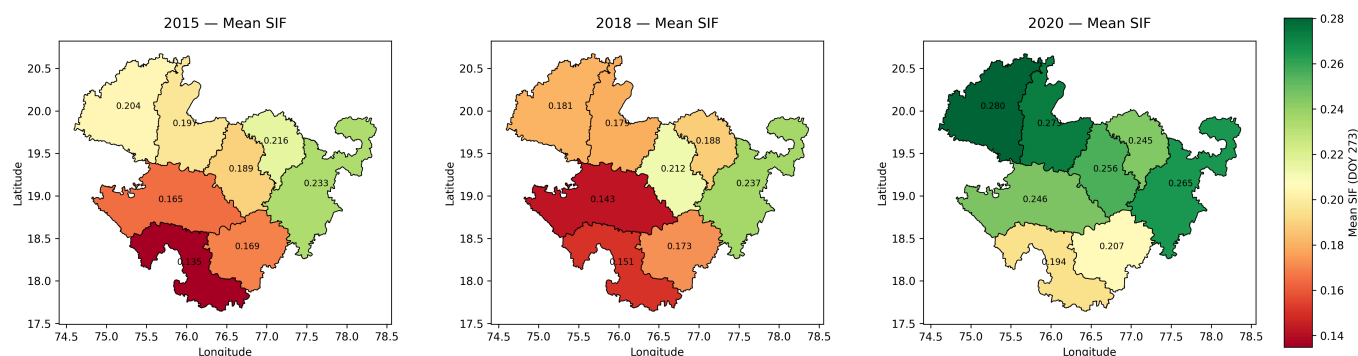


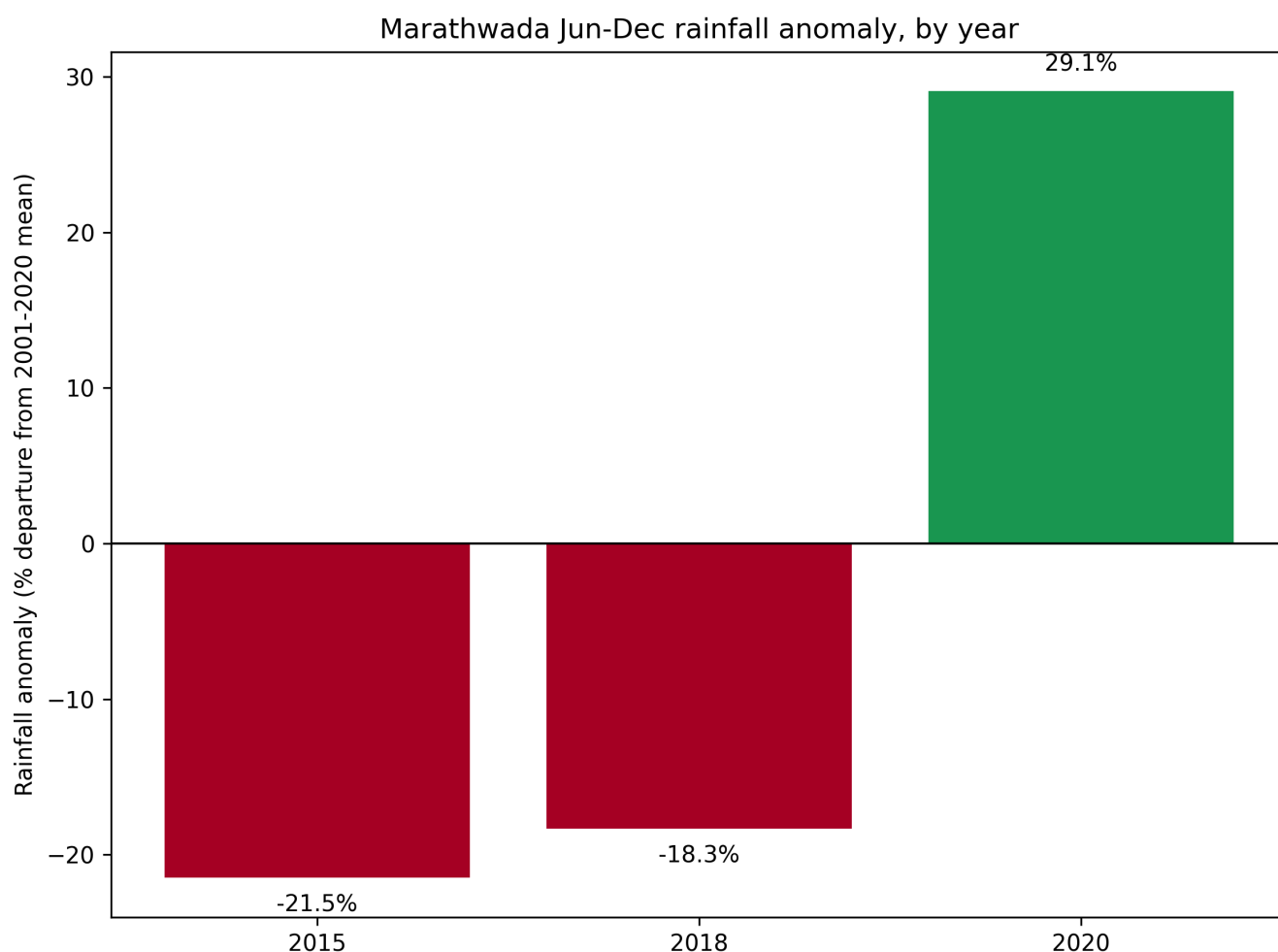
Figure 7. Spatial SIF distribution, day-of-year 273, 2015/2018/2020, masked to the precise Marathwada boundary.

Marathwada mean SIF by district (DOY 273)

**Figure 8.** Mean SIF by district, day-of-year 273, 2015/2018/2020.

Both the pixel-level and district-level spatial views show a consistent low-SIF zone concentrated in the western and southwestern districts (Aurangabad, Bid, Latur, Osmanabad) during both drought years, with Osmanabad recording the lowest district-level mean SIF in all three years studied (0.135, 0.151, 0.194 respectively). Nanded recorded the highest district-level mean SIF in all three years (0.233, 0.237, 0.265), with the smallest relative decline between drought and normal years.

4.4 Rainfall Validation

**Figure 9.** Regional rainfall anomaly (% departure from 2001–2020 climatological mean), by year.

Against a twenty-year climatological mean of 826.4 mm ($\sigma = 158.8$ mm) for the June–December window, 2015 recorded 648.9 mm (−21.5%), 2018 recorded 674.8 mm (−18.3%), and 2020 recorded 1066.6 mm (+29.1%). This independently confirms the drought/normal classification used throughout this study via measured precipitation deficit rather than assumption.

Marathwada rainfall anomaly by district (% departure from regional 20-yr normal)

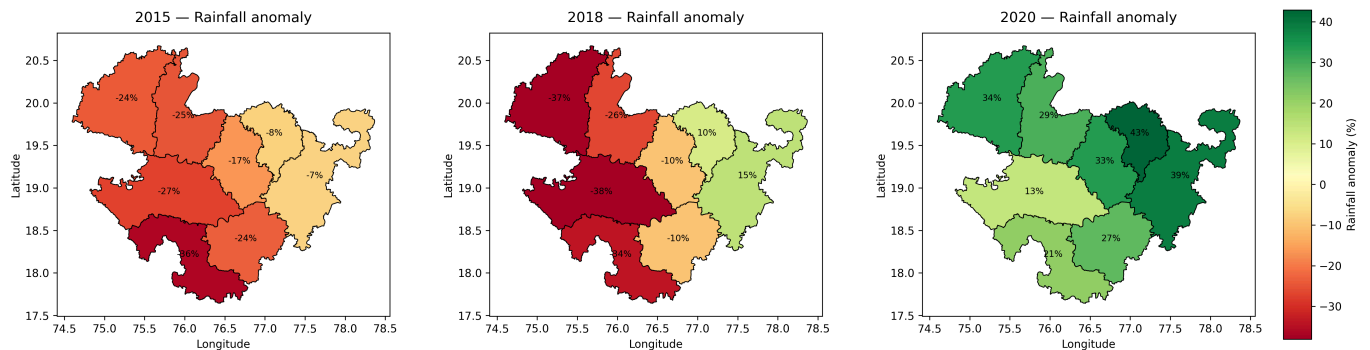


Figure 10. Rainfall anomaly by district (% departure from regional 20-year normal), 2015/2018/2020.

District-level rainfall anomaly reveals that the regional deficit was not spatially uniform. In 2018, Aurangabad (−37.2%) and Bid (−38.1%) recorded the largest deficits, while Hingoli (+10.4%) and Nanded (+14.6%) recorded rainfall surpluses in the same year, despite the region-wide figure indicating an 18.3% deficit. A comparable, though less pronounced, west-to-east gradient was present in 2015.

4.5 Combined Spatial Correspondence

SIF stress vs rainfall deficit — Marathwada, 2015 / 2018 / 2020

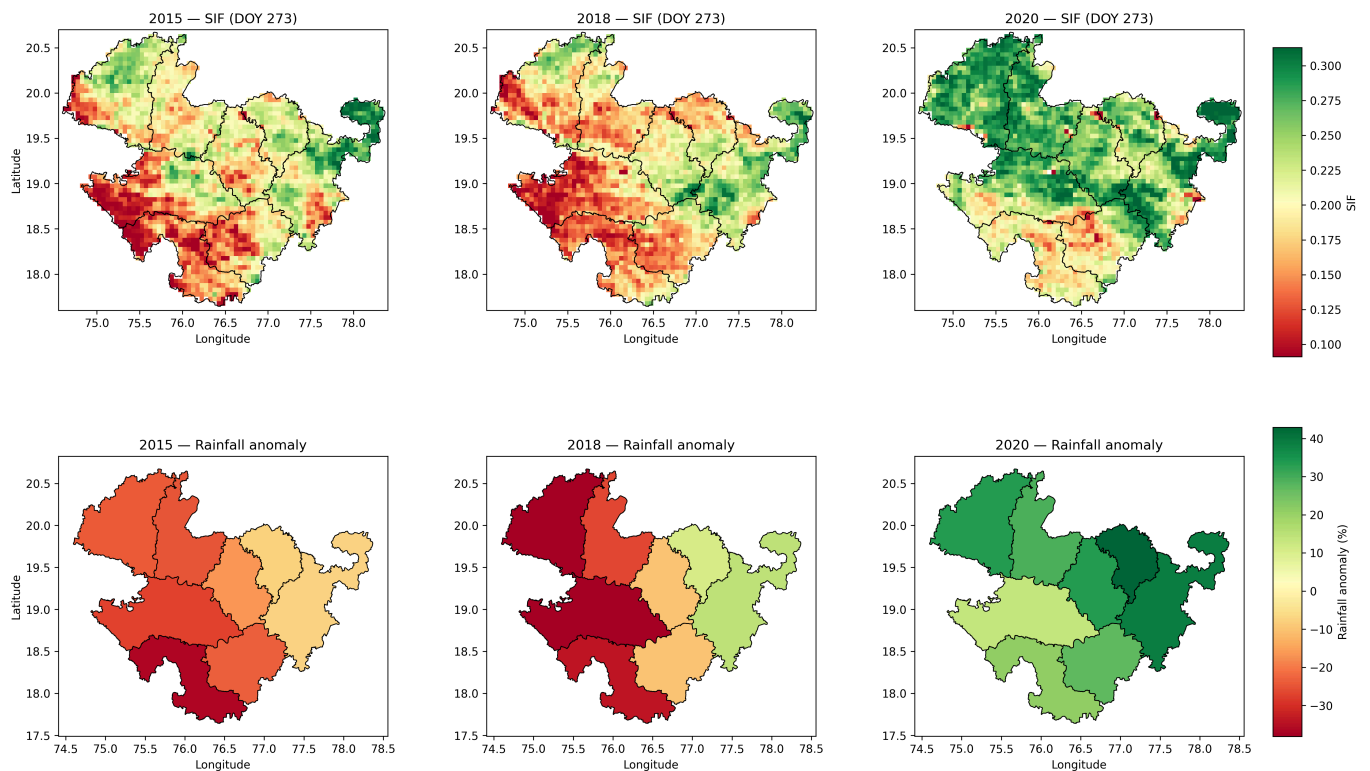


Figure 11. SIF (top row) and rainfall anomaly (bottom row) by district, 2015/2018/2020, presented jointly for direct spatial comparison.

The low-SIF zone identified in Section 4.3 corresponds spatially, in both drought years, with the districts recording the largest rainfall deficits in Section 4.4. Nanded, which recorded the smallest rainfall deficit (or a surplus) in both drought years, also recorded the highest SIF values in the same years.

This visual correspondence was additionally tested statistically across all 24 district-year observations (8 districts $\times$ 3 years). Mean SIF and rainfall anomaly (%) are strongly and significantly correlated: Pearson $r = 0.837$ ($p < 0.001$), and Spearman $\rho = 0.857$ ($p < 0.001$), confirming that the spatial pattern visible in Figure 11 is not an artifact of visual selection. This result carries one caveat: the 24 observations are not fully independent, since the eight districts are geographically adjacent and share regional weather systems, meaning the effective number of independent samples is closer to three (one per study year) than twenty-four. The correlation is reported as calculated, without adjustment for this, and is treated as strong corroborating evidence rather than a fully independent statistical confirmation.

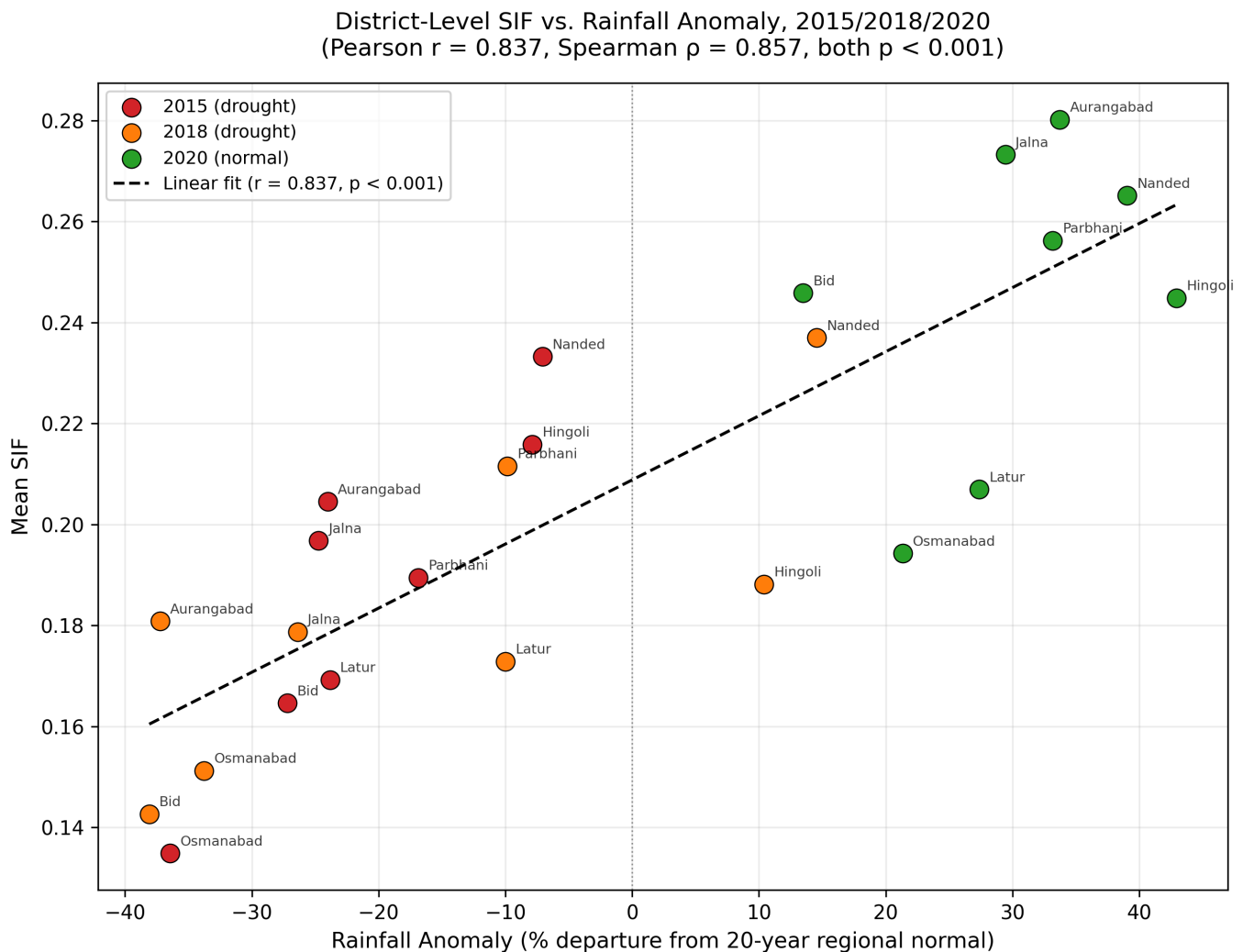


Figure 12. District-level mean SIF plotted against rainfall anomaly (%), all 24 district-year observations (8 districts $\times$ 3 years), with a linear fit (Pearson $r = 0.837$, $p < 0.001$).

4.6 Cross-Correlation Robustness Check

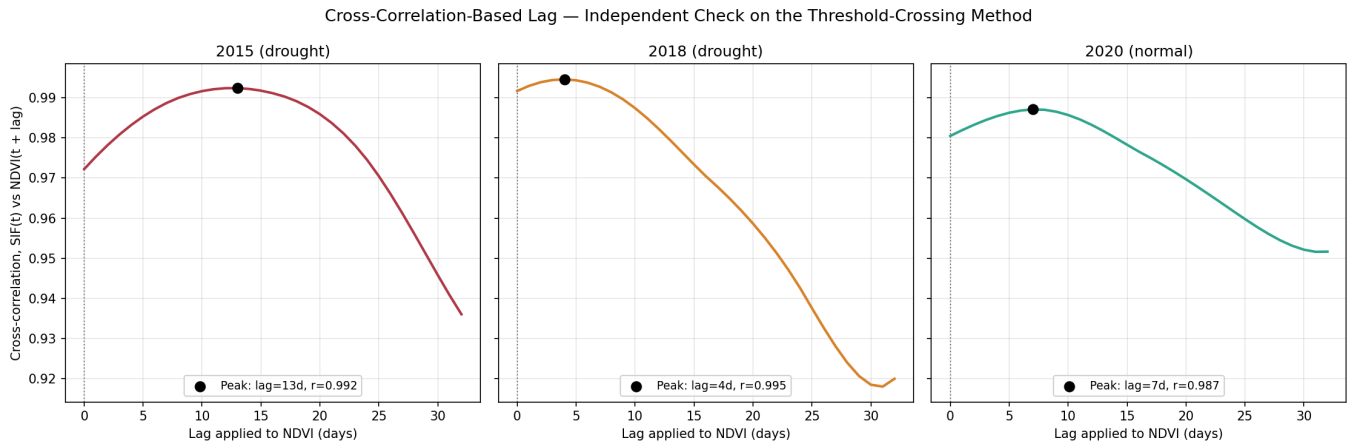


Figure 13. Cross-correlation between SIF and NDVI as a function of the lag applied to NDVI, by year, restricted to lags 0–32 days (N/4 of the 129-day decline window). Black markers show the correlation-maximizing lag for each year.

The cross-correlation-maximizing lag was 13 days (2015), 4 days (2018), and 7 days (2020), each an interior maximum well clear of the search boundary (Figure 13), with correlation values of 0.99, 0.99, and 0.99 respectively — all comfortably above the zero-lag correlation of 0.97, 0.99, and 0.98. Every year's optimal lag is positive, independently confirming, via a wholly different method than Section 4.2, that SIF's decline leads NDVI's decline in all three years — the study's central finding (H1) is robust to a change of lag-estimation method.

The specific *ranking* of lag magnitude between years, however, is not robust to this change of method. The threshold-crossing method (Section 4.2) found the normal year (2020) had the largest mean lag (28.7 days) and 2018 the smallest (6.9 days), with 2015 intermediate (24.3 days). The cross-correlation method instead ranks 2015 highest (13 days), 2020 intermediate (7 days), and 2018 lowest (4 days) — agreeing that 2018 has the smallest lag under both methods, but disagreeing on whether 2015 or 2020 has the largest. This divergence is not a contradiction so much as a consequence of the two methods measuring different things: the threshold-crossing method is most sensitive to the timing of decline *onset*, particularly at high thresholds, while cross-correlation weighs the entire post-peak curve, including the long tail toward the winter minimum. This is reported directly, as an honest finding in its own right, rather than resolved in favor of whichever method's ranking is more convenient — it indicates that this study's specific claim about *which* year shows the largest lag is more methodology-sensitive than the qualitative SIF-leads-NDVI finding itself, and should be read with correspondingly less confidence.

4.7 Uncertainty Quantification Results

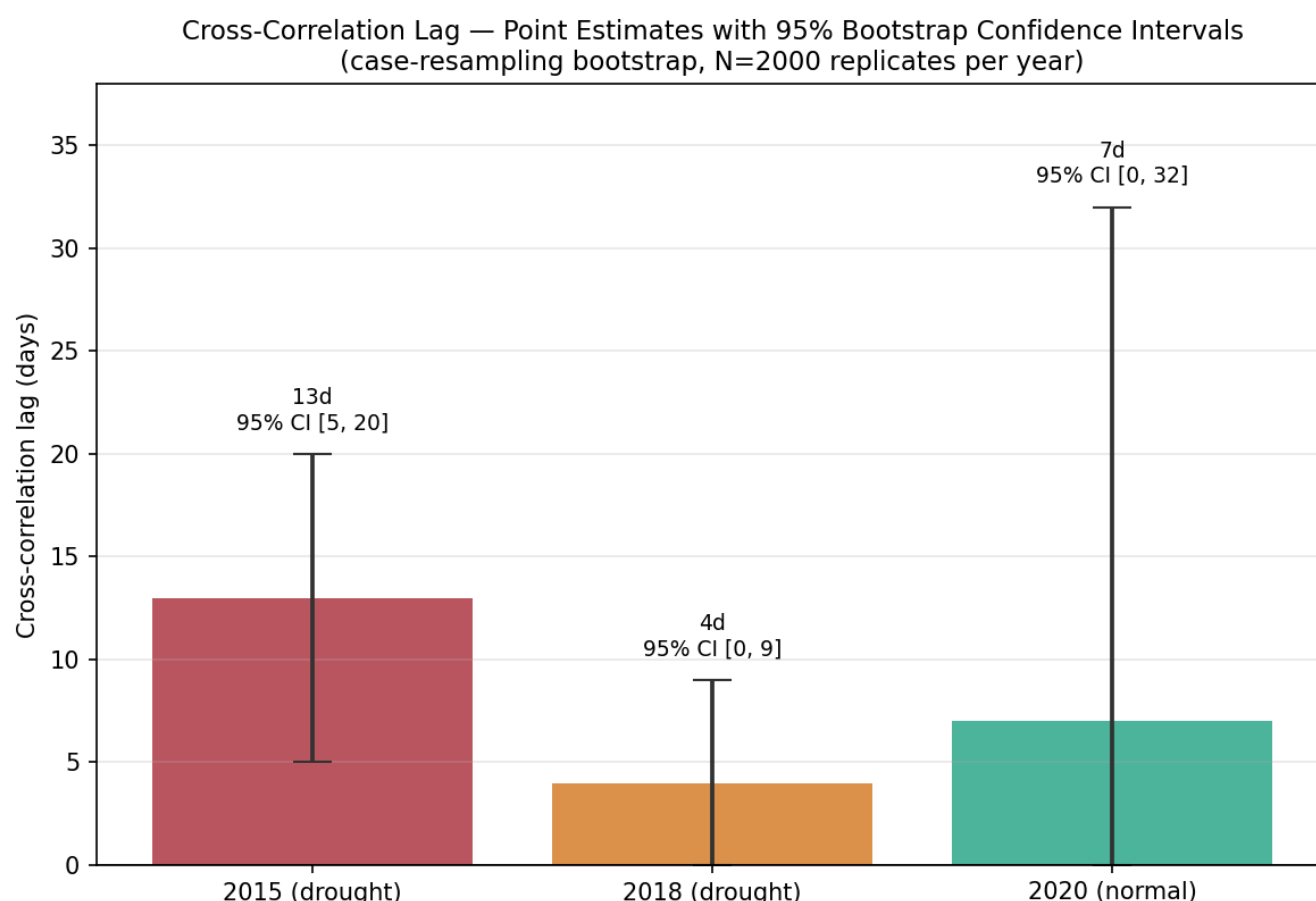


Figure 14. Cross-correlation lag point estimates with 95% bootstrap confidence intervals (case-resampling bootstrap, N = 2,000 replicates per year).

The bootstrap confirms that the *direction* of the lag finding is robust: across every year, the estimated lag was non-negative in 100% of replicates, and strictly positive in 99.9% (2015), 84.5% (2018), and 88.6% (2020) of replicates. The *magnitude*, however, is considerably less certain than the point estimates alone would suggest: the 95% confidence intervals are wide relative to the point estimates (2015: 13 days, CI [5, 20]; 2018: 4 days, CI [0, 9]; 2020: 7 days, CI [0, 32]), and every pairwise comparison between years' intervals overlaps. In other words, this study's finding that SIF leads NDVI is well supported by the bootstrap, but the specific numeric gap between any two years' lag values — including the year-ranking discussed in Section 4.6 — is not statistically distinguishable from noise at this sample size. This is a quantified, stronger version of the caution already given qualitatively in Section 4.6, and reinforces that the exact day-count values reported throughout this paper should be read as approximate rather than precise.

4.8 Spatial Autocorrelation Results

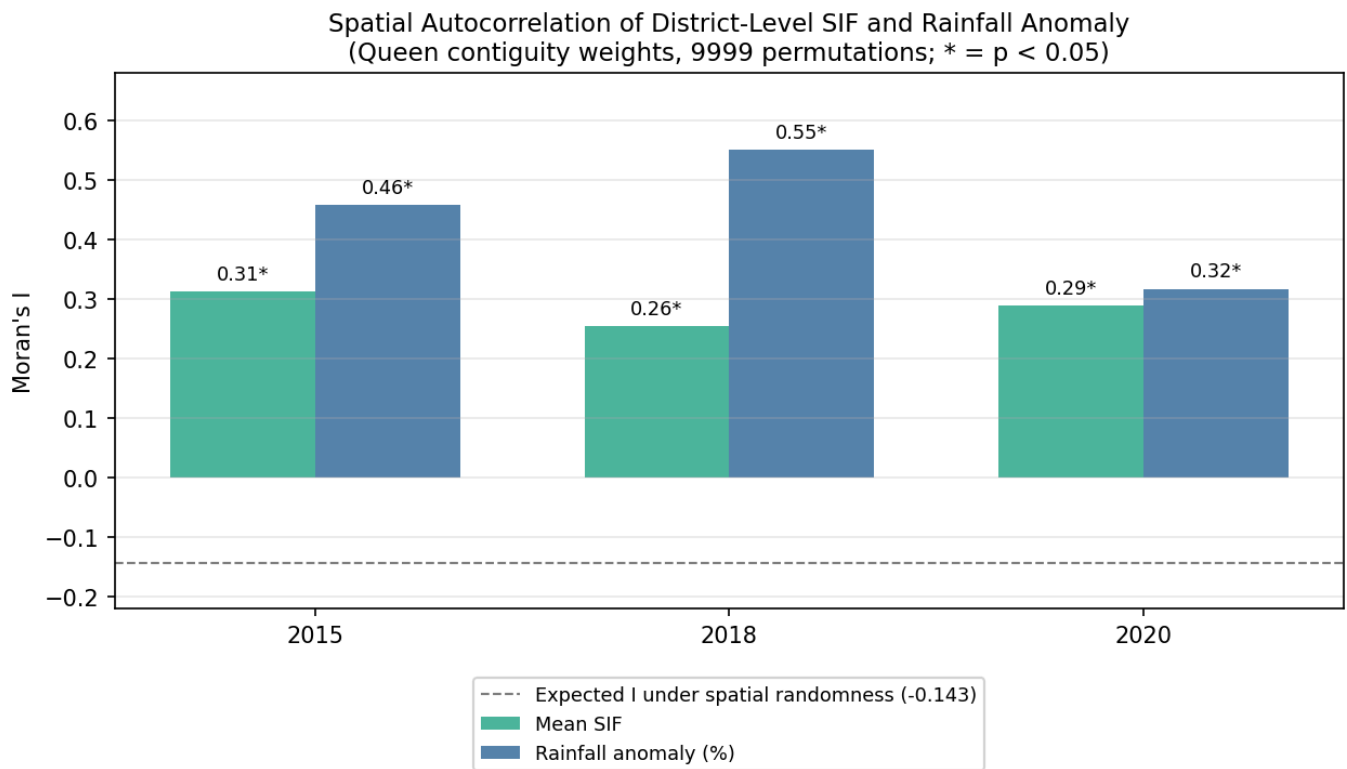


Figure 15. Moran's I for district-level mean SIF and rainfall anomaly, by year, with permutation-based significance markers (* = $p < 0.05$, 9,999 permutations).

All six year $\times$ variable combinations (three years, two variables) showed statistically significant positive spatial autocorrelation (Moran's I ranging from 0.26 to 0.55, all $p < 0.05$), well above the -0.14 value expected under complete spatial randomness for this configuration. This confirms quantitatively, rather than only qualitatively, that neighboring districts' SIF and rainfall values are more similar to one another than chance would predict — validating this paper's existing caveat (Section 4.5) that the effective sample size behind the district-level correlation is closer to the number of years ($n = 3$) than the number of district-year rows ($n = 24$). This does not invalidate the correlation as corroborating evidence — the underlying spatial pattern (Section 4.3, Figures 7–8) is real, physically coherent, and independently cross-validated against rainfall data — but it does mean the correlation's p-value should not be read as though drawn from 24 independent observations.

4.9 Comparison Against the Official Drought-Declaration Timeline (RQ3)

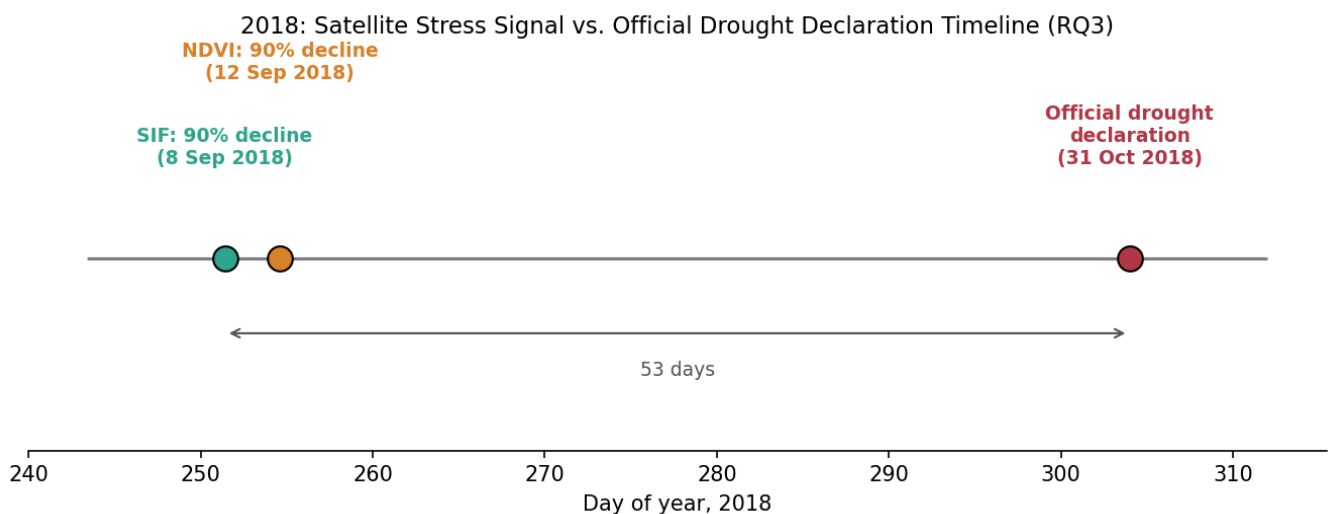


Figure 16. Satellite stress signal versus official drought-declaration timeline, 2018.

Using the 90% decline threshold from Section 4.2 (the most conservative, earliest-triggering threshold, and the one most representative of how an early-warning system would realistically be used), SIF crossed this threshold on 8 September 2018 and NDVI on 12 September 2018 — 53 and 49 days respectively before the Maharashtra government's 31 October 2018 official declaration. Two findings follow from this, and are worth separating. First, the SIF-versus-NDVI advantage that is this paper's central finding is small in absolute terms at this threshold (three to four days) relative to the much larger gap between either satellite indicator and the official declaration (49–53 days, roughly seven weeks). Second, the practical policy case made in Section 1.1 is better read as a case for satellite-based monitoring in general — of which SIF is proposed as a specific, physiologically earlier refinement — rather than a claim that SIF's few-day edge over NDVI is, by itself, the dominant source of potential improvement in payout timing. PMFBY loss assessment already incorporates satellite imagery (Ministry of Agriculture & Farmers Welfare, 2016; a structural reliance on satellite-based area-yield assessment that persists in the scheme as of recent reporting — Down To Earth, 2025), suggesting the operational infrastructure for a satellite-triggered earlier warning already exists in principle. The 2018 comparison here suggests the larger, more immediately actionable gap may be between satellite-based signals of any kind and the current declaration process's timeline, with SIF's specific physiological lead over NDVI as an additional, smaller refinement on top of that larger gap — a more precisely scoped version of this study's policy argument than could be stated before this comparison was run.

5. Discussion

The consistent SIF-leads-NDVI decline pattern observed across all three study years, independent of drought classification, supports the premise that SIF is a more temporally responsive indicator of the onset of physiological change than NDVI (RQ1, supporting H1 in its general form) — a finding that, per Section 4.6, holds under both the threshold-crossing method and an independently computed cross-correlation lag, strengthening confidence in this specific qualitative conclusion beyond what either method alone would support. However, the specific hypothesis that drought conditions amplify this lag (H3) is not supported: the two drought years produced a smaller average lag than the normal year under the threshold-crossing method, and differed substantially from one another. This indicates that the size of the SIF–NDVI lag, at least in this dataset, is not simply a function of drought severity as measured by rainfall deficit — a limitation discussed further below. The precise year-to-year ranking underlying this specific comparison is itself methodology-sensitive (Section 4.6), which further cautions against over-reading the exact magnitude of any single year's lag value.

The magnitude of this discrepancy is itself worth discussing rather than only reporting: 2015 and 2018 recorded broadly similar rainfall deficits (–21.5% and –18.3% respectively, a difference of only about three percentage points), yet their SIF–NDVI lags differed by roughly a factor of three and a half (24.3 versus 6.9 days). If lag size were driven primarily by the magnitude of the seasonal rainfall deficit, these two years would be expected to behave more similarly than they do. Several candidate explanations, none of which this study is able to test directly with the data collected, may account for this divergence. First, the two years may have differed in the timing of rainfall deficit within the season rather than only its total magnitude — a shortfall concentrated early in the monsoon, affecting crop establishment, could plausibly produce different post-peak decline dynamics than one concentrated later, closer to the harvest window that the lag metric specifically measures. Second, the coarser native temporal resolution of NDVI (16 days) relative to SIF (8 days), combined with linear-interpolation-based threshold-crossing estimation, could introduce differential dating uncertainty between years if the rate of post-peak decline itself varied — a steep, rapid decline is more sensitive to

interpolation error than a gradual one. Third, inter-annual differences in dominant Kharif crop choice or sowing calendar within the same districts, which this study did not track, could alter the phenological timing of both indices independently of total rainfall deficit, since different crops exhibit different canopy structures and stress responses. Distinguishing among these explanations would require crop-type and sowing-date records this study did not acquire, and this divergence is reported as an open, unresolved finding rather than one this study is able to adjudicate.

The spatial analysis, combined with independent rainfall validation, provides evidence in support of RQ2: the SIF–NDVI relationship, and the underlying drought stress it reflects, is not spatially uniform across Marathwada. The west-to-east gradient present in both the SIF and rainfall datasets, particularly pronounced in 2018, suggests that region-wide averages used elsewhere in this study can obscure meaningful sub-regional variation.

The bootstrap and spatial-autocorrelation diagnostics added in Sections 4.7–4.8 change how confidently the two headline numeric results should be read, without changing the qualitative conclusions themselves. The bootstrap shows that this study's central qualitative claim — SIF leads NDVI — survives resampling essentially unchanged, while the specific day-count lag values and their year-ranking do not; readers should treat the latter as illustrative rather than precise. Similarly, Moran's I confirms that the district-level SIF-rainfall correlation, while a real and physically coherent pattern, is built from spatially clustered rather than independent observations, and its formal p -value should be discounted accordingly even though the underlying spatial correspondence itself is not in question. Neither diagnostic overturns this study's findings; both make explicit, with a number rather than only a caveat, exactly how much confidence those findings can support.

The comparison in Section 4.9 against the one available official drought-declaration date (2018) refines this study's policy framing. The original motivation (Section 1.1) treats a faster satellite-based indicator as a route to faster PMFBY payouts; the 2018 comparison shows the practically dominant gap is not primarily SIF versus NDVI (a matter of days) but satellite-based monitoring of either kind versus the current declaration and payout process (a matter of roughly seven weeks). This does not weaken the case for investigating SIF further — a physiologically earlier indicator remains worth pursuing on its own terms, and RQ1's finding stands independent of this comparison — but it does mean this study's policy argument is more accurately stated as "satellite monitoring generally, refined by SIF specifically" rather than "SIF specifically" being the primary lever on payout timing.

6. Limitations

- The GOSIF product used for SIF in this study is not a direct satellite retrieval but a statistically modeled reconstruction that uses MODIS reflectance data, alongside OCO-2 SIF soundings and meteorological reanalysis, to produce continuous spatiotemporal coverage. Because MODIS reflectance is also the underlying data source for the NDVI product used for comparison, the two variables are not built from fully independent measurements at the input-data level. This is an inherent property of the most widely available continuous SIF product, not a design choice specific to this study, but it means the magnitude of the SIF–NDVI lag reported in Section 4.2 cannot be fully attributed to independent physical signals, and this study does not attempt to quantify or correct for the resulting modeling dependency.
- The sample size (three years: two drought, one normal) is small, and the two drought years differ substantially from one another, limiting generalizability of any drought/normal comparison.
- The spatial correspondence between rainfall deficit and SIF stress (Section 4.5) was tested via Pearson and Spearman correlation ($r = 0.837$, $\rho = 0.857$, both $p < 0.001$) rather than left as a purely visual

comparison; however, with only three independent study years and eight geographically adjacent districts, the 24 district-year observations are not fully independent, and this correlation should be read as strong corroborating evidence rather than a formally independent statistical confirmation. This is no longer only a qualitative caveat: Section 4.8 quantifies it directly via Moran's I, which is significantly positive ($p < 0.05$) for both SIF and rainfall anomaly in all three years.

- The low-SIF zones identified have not been validated against ground-level crop-stress or drought-impact reporting for the districts concerned.
- District-level rainfall anomaly was computed relative to a single region-wide climatological baseline rather than a per-district climatology.
- Section 4.9 compares SIF- and NDVI-based stress timing against the one well-documented official drought-declaration date located during external review (Maharashtra, 31 October 2018). A comparably verifiable single declaration date for 2015 could not be located in the time available, so this comparison remains partial (one of three study years) rather than complete, and should not be read as a general finding about all study years.
- The Google Earth Engine queries used to acquire the NDVI, land-cover mask, and rainfall data (Sections 3.1–3.2) were originally run interactively in Earth Engine's Code Editor rather than as checked-in scripts. Following external review, a Python translation of this acquisition step (`src/acquisition/gee_data_acquisition.py`, using `earthengine-api/geemap`) has since been added to the repository and is checked into version control. This script has not itself been executed or re-verified against the original interactive output, since doing so requires an authenticated Earth Engine account not available in this project's automated review environment — so it should be read as a substantial reproducibility improvement over prose-only methodology, not as a fully re-verified replacement for the original extraction. The GOSIF clipping and all subsequent processing steps remain fully scripted, executed, and reproducible as before.
- The specific numeric lag values reported in Section 4.2, and particularly the ranking of which study year shows the largest lag, are sensitive to the choice of lag-estimation method: the cross-correlation robustness check (Section 4.6) confirms the direction of the core finding (SIF leads NDVI in all three years) but does not reproduce the same inter-annual ranking as the threshold-crossing method. Section 4.7's bootstrap analysis quantifies this directly: every pairwise between-year comparison of cross-correlation lag estimates has overlapping 95% confidence intervals, meaning the exact ranking is not statistically distinguishable from noise at this sample size. Readers should treat the qualitative SIF-leads-NDVI finding as the robust result of this study, and the exact day-count lag values as method-dependent, wide-uncertainty estimates rather than precise, method-independent quantities.

7. Conclusion

This study finds consistent evidence that Solar-Induced Fluorescence registers the onset of post-peak seasonal vegetation decline earlier than NDVI across all years studied in Marathwada, Maharashtra, supporting SIF's physical basis as a more temporally responsive stress indicator — a finding corroborated by an independent cross-correlation method and shown, via bootstrap resampling, to be robust in direction even where exact magnitudes are not. It does not find evidence that this lag is amplified specifically by drought conditions, and identifies substantial inter-annual and intra-regional variation that a simple drought/normal binary does not capture. Independent rainfall validation and spatial cross-referencing between SIF and precipitation data lend physical coherence to the district-level findings — a correspondence confirmed statistically (Pearson $r = 0.837$, Spearman $\rho = 0.857$, both $p < 0.001$), and further characterized, rather than merely caveated, via a Moran's I spatial-autocorrelation diagnostic — while a comparison against the one available official drought-declaration date (2018) refines this study's policy framing toward satellite monitoring in general, with SIF as a specific

physiological refinement, rather than positioning SIF's few-day edge over NDVI as the dominant lever on payout timing. Several limitations — sample size, absence of ground validation, the spatial non-independence underlying the district-level correlation, and a Google Earth Engine acquisition step that is now scripted but not yet re-executed — are reported directly rather than resolved beyond what the available data supports.

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